African elephants address one another with individually specific name-like calls

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Abstract

Personal names are a universal feature of human language, yet few analogues exist in other species. While dolphins and parrots address conspecifics by imitating the calls of the addressee, human names are not imitations of the sounds typically made by the named individual. Labelling objects or individuals without relying on imitation of the sounds made by the referent radically expands the expressive power of language. Thus, if non-imitative name analogues were found in other species, this could have important implications for our understanding of language evolution. Here we present evidence that wild African elephants address one another with individually specific calls, probably without relying on imitation of the receiver. We used machine learning to demonstrate that the receiver of a call could be predicted from the call's acoustic structure, regardless of how similar the call was to the receiver's vocalizations. Moreover, elephants differentially responded to playbacks of calls originally addressed to them relative to calls addressed to a different individual. Our findings offer evidence for individual addressing of conspecifics in elephants. They further suggest that, unlike other non-human animals, elephants probably do not rely on imitation of the receiver's calls to address one another.